AI-Powered Resume Screening & Candidate Ranking System

Natural Language Processing (NLP) & Supervised Skill Alignment Pipeline

Internship Task: Future Interns Machine Learning Internship — Task 3

Project Identifier: FUTURE_ML_O3

Date of Submission: August 2026

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1. Executive Summary & Business Context

In modern recruitment, Human Resource departments are overwhelmed by the sheer volume of resumes submitted for open positions. Manual resume screening is time-consuming, prone to cognitive fatigue, and introduces subjective biases. To optimize recruitment workflows, organizations require decision-support tools that quickly filter, analyze, and rank applicants according to job-specific requirements.

This project, developed under the **Future Interns Machine Learning Internship (Task 3)**, implements an end-to-end NLP and Machine Learning pipeline. The system processes raw candidate profiles, performs keyword and semantic text similarity analysis, matches technical skill sets, analyzes missing qualifications, and ranks applicants to help recruiters prioritize high-potential candidates.

2. Technical Architecture & Methodology

The application architecture consists of five core stages:

- **Data Extraction:** Ingestion of the Kaggle Resume Dataset (2,400+ resumes scraped from livecareer.com) and anonymization of profiles with CAND_XXX IDs.
- **NLP Preprocessing:** Lowercasing, removal of HTML tags, URLs, email addresses, phone numbers, and custom tokenization rules designed to preserve critical programming tags (e.g., C++, .NET, C#) that standard tokenizers corrupt.
- **Supervised Skill Extraction:** Regex-based dictionary lookup using a configurable technical skill dictionary of 40+ concepts grouped into distinct categories (Databases, Programming, BI, Cloud, etc.).
- **TF-IDF Vectorization & Similarity:** A TfidfVectorizer fits on the preprocessed corpus and transforms resumes and job descriptions. Textual Cosine Similarity is calculated as a baseline matching score.
- **Transparent Hybrid Scoring:** Candidates are ranked using a weighted fit score: 60% TF-IDF Cosine Similarity and 40% Exact Skill Match percentage.

3. Scoring & Skill Matching Formulas

Rather than relying on closed-source algorithms, this project uses transparent and explainable equations:

Skill Match Percentage:
$$\text{Skill Match \%} = \left(\frac{\text{Number of Matched Required Skills}}{\text{Total Number of Required Skills}} \right) \times 100$$

Final Candidate Score:
$$\text{Final Score} = (0.60 \times \text{Cosine Similarity \%}) + (0.40 \times \text{Skill Match \%})$$

4. Ethical Considerations & Limitations

Recruiter Decision-Support: The screening system is designed explicitly as a decision-support assistant. It is **not** an automated decision-making system. The final hiring and evaluation steps must remain under human control to prevent bias and ensure compliance with employment legislation.

Biases & Fairness: Keyword and TF-IDF similarity algorithms rank resumes based on wording. Candidates who describe their skills using different terminology or who have non-traditional career paths might receive lower scores, even if highly qualified. Hiring managers must ensure human oversight to verify rankings and avoid discriminating against non-standard resume designs.

System Limitations: The current model performs lexical (exact keyword) matching and bag-of-words similarity. It does not fully capture semantic context (e.g., distinguishing 'learning machine learning' from 'managing a machine learning team'). Experience depth and education credentials are also not parsed in this prototype.

5. Expected Project Outcomes

Upon full pipeline execution, the system generates: ranked candidate sheets (`ranked_candidates.csv`), visual skill gap distribution analytics (`skill_gap.png`, `top_skills.png`), and launches a web dashboard utilizing Streamlit. The dashboard allows recruiters to adjust algorithm weights in real time, paste job descriptions dynamically, upload PDF resumes, and read contextual candidate fit justifications.

This project successfully combines foundational natural language processing, data analysis, and visual business reporting, making it a strong submission for the Machine Learning Internship portfolio.